

Paper 3 Theorem 2.2 — Statement Amendment

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Source checked

This note checks the original source, not the repository paraphrase:

- Le Chen, Ian Ruau, and Wenxian Shen, *Chemotaxis models with signal-dependent sensitivity and a logistic-type source, II: Persistence and stabilization*, arXiv:2604.02599v1, 3 April 2026.
- Theorem 2.2 there is titled “Linear stability and instability.”

Theorem 2.2 concerns the positive constant equilibrium (u^*, v^*) , $u^* = (a/b)^{1/\alpha}$, of the bounded-domain Neumann problem. Part (1) states, for $a, b > 0$:

1. if $\chi_0 < \chi_{a,b,\beta}^*(u^*)$ the equilibrium is **linearly stable**;
2. if $\chi_0 > \chi^*$ it is **linearly unstable**;
3. moreover, in the stable regime there exist $C, \delta, \lambda > 0$ such that every u_0 satisfying the initial-data condition (1.8) **and** a smallness condition obeys the exponential estimate (2.12) **for all $t \geq 0$** in the C^1 norm.

Part (2) is the corresponding minimal-model statement ($a = b = 0$), with the mass constraint added.

The initial-data hypothesis (1.8) requires only u_0 in $C(\bar{\Omega})$ (continuous, positive). Proposition 1.1 supplies convergence to u_0 in L^∞ as $t \rightarrow 0+$, and classical spatial regularity only for **positive** time. There is **no** C^1 , Sobolev, or fractional-power smallness hypothesis on u_0 in the statement of Theorem 2.2.

The over-statement in the published (2.12)

The printed nonlinear clause asserts, for data small only in L^∞ , a C^1 exponential estimate that holds **at every $t \geq 0$, including $t = 0$** :

$$\|u(t) - u^*\|_{C^1} \leq C e^{-\delta t} (\dots), \quad \text{for all } t \geq 0.$$

This clause is not well-posed under the theorem’s own initial-data hypothesis:

- u_0 is only required to lie in $C(\bar{\Omega})$, so $\|u_0 - u^*\|_{C^1}$ need not be finite; the $t = 0$ instance of (2.12) can be meaningless.
- Even restricting to C^1 data, an L^∞ -ball contains functions of arbitrarily large C^1 norm, so no single constant C can control the $t = 0$ (or $t \rightarrow 0+$) C^1 size uniformly over the admitted class.

Concretely, on the unit interval $(0,1)$ with Neumann conditions take

$$u_{0,N}(x) = u^* + N^{-1/2} \cos(N\pi x).$$

For large N this satisfies (1.8), is positive, has $\|u_{0,N} - u^*\|_{\infty} = N^{-1/2} \rightarrow 0$ (and, in the minimal model, exact mass u^*), yet $\|u_{0,N} - u^*\|_{C^1} = N^{1/2} \pi \rightarrow \infty$. So no uniform-in-data $t = 0$ C^1 estimate of the form (2.12) can hold. The smallness is L^∞ ; the conclusion is measured in C^1 ; the two are unrelated at $t = 0$.

What the published proof actually establishes

Section 5.1 proves the result in two distinct stages, both correct:

- **Stage A (strong-norm, all-time).** For data small in the fractional-power space $X^\alpha = D((I - L)^\alpha)$ (which, in one dimension, embeds into C^1 for $\alpha > 3/4$), the standard sectorial nonlinear-stability theorem (Henry, *Geometric Theory of Semilinear Parabolic Equations*, Ch. 5-6) gives exponential decay in X^α for all $t \geq 0$. This is instantaneous and correct.
- **Stage B (L^∞ -small, eventual).** For data small only in L^∞ , the proof first evolves to a **positive** time T_0 , at which the analytic-semigroup smoothing places the solution in a small X^α neighbourhood, and then applies Stage A. This yields exponential decay for $t \geq T_0$ — an *eventual* estimate, not a $t = 0$ estimate.

Thus the mechanism proves (A) an all-time estimate in the strong norm and (B) an eventual estimate in C^1 for L^∞ -small data. It does **not** prove the printed (2.12) at $t = 0$ from L^∞ data; that conjunction is the over-statement.

Correct statement

The faithful correction splits the printed clause into the two statements the proof supports:

1. **Strong-norm, all-time:** there is α in $(3/4, 1)$ and $\rho, C, \delta > 0$ such that if $\|u_0 - u^*\|_{X^\alpha} \leq \rho$ then $\|u(t) - u^*\|_{X^\alpha} \leq C e^{-\delta t} \|u_0 - u^*\|_{X^\alpha}$ for all $t \geq 0$.
2. **L^∞ -small, eventual:** there is $\epsilon, C', \delta' > 0$ and, for each admissible datum, a time $t_0 = t_0(u_0) > 0$ such that if $\|u_0 - u^*\|_{\infty} \leq \epsilon$ then $\|u(t) - u^*\|_{C^1} \leq C' e^{-\delta'(t - t_0)}$ for all $t \geq t_0$.

The linear stability/instability dichotomy of Theorem 2.2 is unaffected and correct; only the nonlinear clause (2.12) needs the $t = 0$ -versus-eventual and L^∞ -versus- X^α distinctions made explicit.

A companion parameter-fidelity point

The sufficient stability bound $\kappa < (\sqrt{\mu} + \sqrt{a \alpha})^2$ (with $\kappa = \chi_0 \gamma \nu u^{\gamma} (1 + v)^{-\beta}$) is the *continuous* minimum of $(\lambda + \mu)(\lambda + a \alpha) / \lambda$ over $\lambda > 0$. On a bounded domain the Neumann spectrum is discrete ($\lambda_n = n^2 \pi^2$ on $(0,1)$), so the exact stable half of Theorem 2.2 is governed by the **discrete** infimum $\min_{n \geq 1} \sigma_n < 0$, which the continuous bound only implies (and can be strictly stronger than needed). A faithful formalization should clear the discrete mode condition, not stop at the continuous sufficient bound.

Lean amendment

The repository records two formal obstructions to the literal clause:

`Paper3.not_SectorialLocalExponentialRaw_constant_c1Distance`
`(ShenWork/Paper3/Statements.lean)`

proves that the raw sectorial local-exponential branch is false when the C^1 distance is unrelated to the dynamics: a constantly-1 C^1 distance would force $2 \leq C \exp(-\text{rate } t)$ for all t , impossible as the right side tends to 0. Independently,

`ShenWork/Paper3/IntervalDomainSectorialCorrectedObstruction.lean`

records a **zero-time** obstruction: `InitialTrace` controls only the $t \rightarrow 0+$ deleted-limit, not the stored slice $u(0)$, so a global solution can be re-anchored to an arbitrarily large constant slice at $t = 0$ without changing global classical solvability or the initial trace — hence no all-time $t = 0$ C^1 bound can hold. Both are sorry-free.

The corrected targets used in the repository are the eventual, equilibrium-specific, full-mode orbit bound

`IntervalDomainSpectralSemigroupOrbitBoundEventualEquilibriumWithoutMass`

(full linearized multiplier $\exp(-d_k t)$, $d_k = a \alpha + \lambda_k - \kappa \lambda_k / (\lambda_k + \mu)$ for every mode k including $k = 0$, with a uniform positive delay t_0), together with `LocallyExponentiallyStableFromSup` for the positive equilibrium (no mass constraint, which is needed only in the minimal $a = b = 0$ model). The over-stated `...Raw` frontier — the pure Neumann heat multiplier quantified over all $t \geq 0$ — is not the correct analytic object and is refuted for a genuine C^1 distance from L^∞ data.

Relation to Paper 1 (arXiv:2512.14858) Theorem 1.2

This is the **second** statement-level issue found during formalization of the Chen-Ruau-Shen series. The first, recorded in `paper2_theorem12_statement_amendment.md`, is a genuine *parameter* error: Theorem 1.2 of Part I omits the guard excluding $a > 0$, $b = 0$, a branch in which the solution is provably unbounded ($M'(t) = a M(t)$). The present issue is a *statement over-statement*: the intended

stability result of Part II holds in the corrected two-stage form above, but the printed (2.12) — a $\tau = 0$ C^1 estimate from L^∞ -small data — is not well-posed as written.